
EE 540: Wide Band Gap Devices in Power Electronics Applications

Course Code	EE 540
Course Name	Wide Band Gap Devices in Power Electronics Applications
L-T-P-C	3-0-0-3
Prerequisites	Power Electronics (EE 309/EE527/ EE504) or Power Semiconductor Devices (EE526) or equivalent
Intended for	UG/MS/MTech/PhD
Distribution	Elective: B. Tech III and IV year /M.S./ M.Tech / PhD
Approval	recommended by the 35th BoA (Aug 2020)

Course Contents

- **Wide band gap devices:** [4 Lectures]
 - Introduction of wide band-gap
 - Vertical and lateral structures of wide band-gap devices
 - Different types of the wide band gap devices
 - Advantages of wide band-gap semiconductors
 - Challenges in designing converters with wide band-gap devices
- **Switching characteristics:** [4 Lectures]
 - Turn-on and Turn-off characteristics of the device
 - Hard switching loss analysis
 - Double pulse test set-up
- **Drivers for wide band-gap devices:** [8 Lectures]
 - Gate driver
 - Impact of gate resistance
 - Gate drivers for wide bandgap power devices
 - Transient immunity integrated gate drivers
 - Overcurrent protection, UVLO protection
- **Thermal management of power converters:** [6 Lectures]
 - Thermal modelling
 - Thermal management and reliability
 - Improving the performance with heatsink
- **High frequency design complexity:** [4 Lectures]
 - Effects of parasitic inductance
 - Effects of parasitic capacitance
 - EMI filter design for high frequency power converters
- **PCB designing:** [4 Lectures]
 - High frequency PCB design
 - Conventional power loop design
 - High frequency power loop optimization
 - Single and multi-layer PCBs
 - Separation of power from signal PCB
- **Power density advantages:** [4 Lectures]

- Power density of wide bandgap devices
- High power density power electronics converters
- **Applications of wide bandgap devices:** [8 Lectures]
 - Consumer electronics applications
 - Wireless power transfer applications
 - Electric vehicle applications
 - Renewable energy sources applications
 - Students will carry out one project based on modeling, simulation of any of the wide bandgap device in power electronics applications. This work can be extended with the practical work.

Text Books

1. A. Lidow, J. Strydom, M. D. Rooij, D. Reusch, **GaN Transistors for Efficient Power Conversion**, Wiley, 2014.
2. G. Meneghesso, M. Meneghini, E. Zanoni, **Gallium Nitride-enabled High Frequency and High Efficiency Power Conversion**, Springer International Publishing, 2018.

References

1. Ned Mohan, Tore M. Undeland, William P. Robbins, **Power Electronics**, John Wiley & Sons, 2003.
2. F. Wang, Z. Zhang and E. A. Jones, **Characterization of Wide Bandgap Power Semiconductor Devices**, IET, 2018.
3. L. Umanand and S. R. Bhat, **Design of Magnetic Components for Switched Mode Power Converters**, John Wiley & Sons, 1992.
4. B.J.Baliga, **Gallium Nitride and Silicon Carbide Power Devices**, World Scientific Publishing Company, 2017.
5. E. L. Corradini, D. Maksimovic, P. Mattavelli, R. Zane, **Digital Control of High-Frequency Switched-Mode Power Converters**, Wiley, 2015.